PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville expected 2027
Ph.D. in Economics — Committee: James Lake, Georg Schaur, Matthew Van Essen [[confirm]]

Shandong University 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (950+ citations)

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

Environmental and Energy Economics, Agricultural and Resource Economics, Applied Econometrics, and International Trade.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." With K. Du and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **950+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." With T. Yu et al. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." With T. Yu et al. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." With C. Martinez et al. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake [JMP]
Abstract: Trade-restrictiveness indices summarize a country's tariff schedule as a single welfare-equivalent uniform tariff, but the existing indices assume most-favored-nation tariffs. Since 2018 the largest tariff changes—the US–China tariff war, retaliatory tariffs, and proliferating preferential agreements—have been discriminatory by design. We extend the trade-restrictiveness index to partner-specific tariffs while preserving its welfare interpretation and derive a closed-form decomposition into a "level" component (how much a country restricts trade) and a "discrimination" component (how unevenly it does so). Computing the indices from WITS/Comtrade data for 2017–2024, we find that the MFN-only index systematically understates trade distortions when discrimination is high, and the decomposition separates, for the first time in this framework, the welfare cost of discrimination from the cost of protection. [[confirm / replace with paper abstract]]

Welfare Analysis of a Forest-Residue Sustainable Aviation Fuel Supply Chain under Endogenous Demand and Price Uncertainty with T. Edward Yu

Stochastic Optimisation of Logging Residue–based Biofuel Supply Chains Considering

Feedstock Ash Content and Biofuel Price with T. Yu et al.

Sustainable Aviation Fuel Production from Forest Residues with Ash Content: A Case Study of Nashville Airport with T. Yu et al.

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, Tun-Hsiang Yu, and David Anderson. SSRN 4121953 (2022).

The Impact of Mayoral Endorsements on Presidential Election Outcomes with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*
Abstract: We develop a mean-field game in which local politicians strategically time their endorsements of presidential candidates and the aggregate of those decisions feeds back into voter beliefs. We characterize equilibrium endorsement dynamics in the mean-field limit, derive comparative statics in electoral closeness and mayors' career incentives, and test the model on mayoral endorsement data from Brazil and France. [[confirm]]

WORK IN PROGRESS

Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value with Georg Schaur

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Tennessee 2020 – 2023
USDA NIFA (two projects) and FAA Office of Environment and Energy (ASCENT) funded research on forest-residue biofuel and sustainable aviation fuel (SAF) supply chains in the Southeast U.S.

- Designed cost-minimizing and profit-maximizing supply-chain networks with mixed-integer programming (GAMS); results published in *Biomass and Bioenergy* and *Transportation Research Record.*
- Facility-siting and investment analysis combining econometric models with the IMPLAN regional impact platform; efficiency analysis of facilities and airports with Data Envelopment Analysis.
- Sensitivity and scenario analysis of feedstock quality and market conditions with extended mathematical programming (EMP), supervised machine learning, and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

Other research experience

- *Panel and causal methods:* synthetic-control estimate of the long-run output effect of the 2008 Wenchuan earthquake; GMM estimates of MFP/CFAP payments on U.S. agricultural lending; Hansen panel-threshold models of green innovation and CO_2; mediation analysis of digital finance and emissions for Chinese listed firms.
- *Time series:* threshold VAR of regime switching in commodity-price linkages; VECM and multivariate GARCH of wholesale commodity-price volatility.
- *Machine learning:* drivers of CO_2 emissions from patent data; equity-premium prediction from *NYT* and *WSJ* text.

PRESENTATIONS

Western Regional Science Association, Annual Meeting, Hawaii 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA 2022
Midwest Economic Theory and International Trade Meetings, Virginia Tech Spring 2025

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics — Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development — Summer 2025
ECON 313: Intermediate Macroeconomics — Summer 2026

Teaching Assistant

Ph.D. courses:
ECON 511: Microeconomic Theory — Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics — Spring 2024
Undergraduate and master's courses:
ECON 581: Mathematical Methods in Economics — Summer 2024
AREC 525: Agribusiness Operations Research Methods — Spring 2022, Spring 2023

HONORS AND AWARDS

J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee — 2024
Graduate Assistantship, University of Tennessee — 2020 – present
Outstanding Volunteer, migrant-family tutoring program, Shanghai — [[year]]

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

T. Edward Yu
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University of Tennessee
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Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
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James Lake
Associate Professor of Economics
Department of Economics
University of Tennessee
Email: jlake7@utk.edu